Ashkan Khademi

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Education

Toronto Metropolitan University — Toronto, ON
Master of Engineering in Electrical and Computer Engineering (AI Specialization) — *September 2025 – June 2027*

York University (Lassonde School of Engineering) — Toronto, ON
Bachelor of Science with Honours in Computer Science — *September 2020 – December 2024*

Technical Skills

Languages: Python, Java, JavaScript, SQL, C, HTML, CSS
AI/ML: PyTorch, Hugging Face Transformers, LangGraph, LLMs, NLP, Conformal Prediction, Pandas, NumPy
Frameworks/Libraries: FastAPI, React.js, Node.js, Express.js, Spring, Streamlit, JUnit
Tools/Technologies: Git, Docker, PostgreSQL, MongoDB, SQLite, Qdrant, Supabase, REST APIs, Power Automate

Work Experience

AI Technical Documentation Intern — May 2026 – October 2026
Communications & Power Industries (CPI) — *Georgetown, ON*

- Architected a FastAPI app turning natural-language instructions into LLM-proposed edits to Word documents.
- Implemented an 8-stage validation pipeline filtering unsupported LLM edits before native DOCX changes.
- Secured the app with Microsoft Entra SSO (OAuth 2.0, JWT) and wrote 1,750+ automated regression tests.
- Automated purchase-order intake with Power Automate, AI Builder OCR and GPT extraction into Excel.
- Streamlined order review with confidence thresholds, duplicate detection and discrepancy checks.

Software Engineer Intern — December 2025 – April 2026
TranQuility Inc. — *Toronto, ON*

- Built REST APIs integrating a 7-agent LangGraph workflow with synchronous and SSE streaming responses.
- Applied LLM-based fact extraction to persist cross-session semantic memory in Qdrant and chats in Supabase.
- Collaborated with the engineering team to fix memory-write failures and stabilize background streaming for demos.

Software Developer Intern — February 2025 – September 2025
ReliablyME Inc. — *Toronto, ON*

- Created a Python-based cohort projection model using Pandas and NumPy to forecast revenue growth, churn and conversion across subscription plans.
- Engineered a cohort tracking system to monitor customer transitions across 4 plan tiers, improving retention visibility and enabling monthly CAC and churn reporting.
- Devised models to simulate 48-month trends and forecast churn, MRR and retention from cohort data.

Research Experience

Conformal Prediction for Toxicity Detection — May 2026 – Present
MEng Research Project, Toronto Metropolitan University — *Toronto, ON*

- Developing a post-hoc uncertainty quantification framework using conformal prediction to deliver statistically guaranteed multi-label coverage for frozen NLP toxicity classifiers without model retraining.
- Evaluating 18 detector-dataset pairs across 7 benchmarks with a Python, PyTorch and Hugging Face Transformers pipeline, covering BERT-family models (DistilBERT, XLM-R) and LLMs (Gemini, Qwen3, GPT-OSS).
- Benchmarked against 14 uncertainty baselines with statistical significance testing, achieving 0.902 coverage at a 0.90 target with 21.8% smaller prediction sets than a standard conformal baseline.

Cybersecurity and DPI Anomaly Detection — September 2024 – December 2024
Capstone Project, York University — *Toronto, ON*

- Researched real-time anomaly detection with Beta-VAE and Suricata, improving detection from 0.1% to 77.8%.
- Designed an AI-driven intrusion detection system, analyzing 50,000+ network packets for suspicious activity.
- Integrated live packet monitoring with Scapy, enabling real-time analysis of encrypted and unencrypted traffic.

Projects

AskMyPDF (Python, Streamlit, Sentence-Transformers, NLP)

- Crafted an AI chatbot to extract answers from uploaded PDFs using embeddings, reducing false matches by 80%.
- Processed and indexed 250+ text chunks per document with PyMuPDF and Sentence-Transformer models.
- Enabled uploads of PDF files up to 100 MB with Streamlit, ensuring interactive sub-second response times.